High-fidelity multipartite entanglement creation in non-Hermitian qubits

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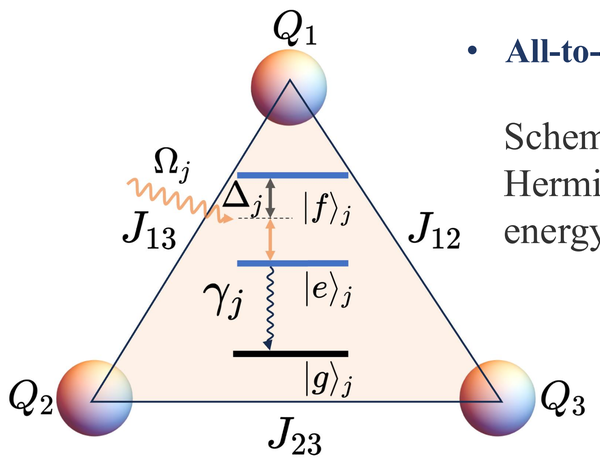
Introduction

Non-Hermitian physics describes open quantum systems, exhibiting unique features like exceptional points (EPs) that enhance sensitivity. We focus on **strong driving/coupling regimes** to achieve **high-fidelity GHZ states**, offering a new route for quantum computation.

Theory



- **All-to-all coupling**

Schematic of non-Hermitian three qubits energy configurations

- **Hamiltonian**

$$\hat{H} = \sum_{j=1}^{3}\left[\left(\Delta_j - \frac{i\gamma_j}{2}\right)\hat{\sigma}_j\hat{\sigma}_j^\dagger + \Omega_j\hat{\sigma}_j^x\right] + \sum_{j=1}^{3}\sum_{k=j+1}^{3} J_{jk}\left(\hat{\sigma}_j^\dagger\hat{\sigma}_k + \hat{\sigma}_j\hat{\sigma}_k^\dagger\right)$$

Detuning between qubits level — Non-Hermitian spontaneous decay — Driving fields on qubits — All-to-all coupling between qubits

Objective

→ Engineer non-Hermitian interactions to robustly generate high fidelity GHZ state / entangled state, even under dissipation.

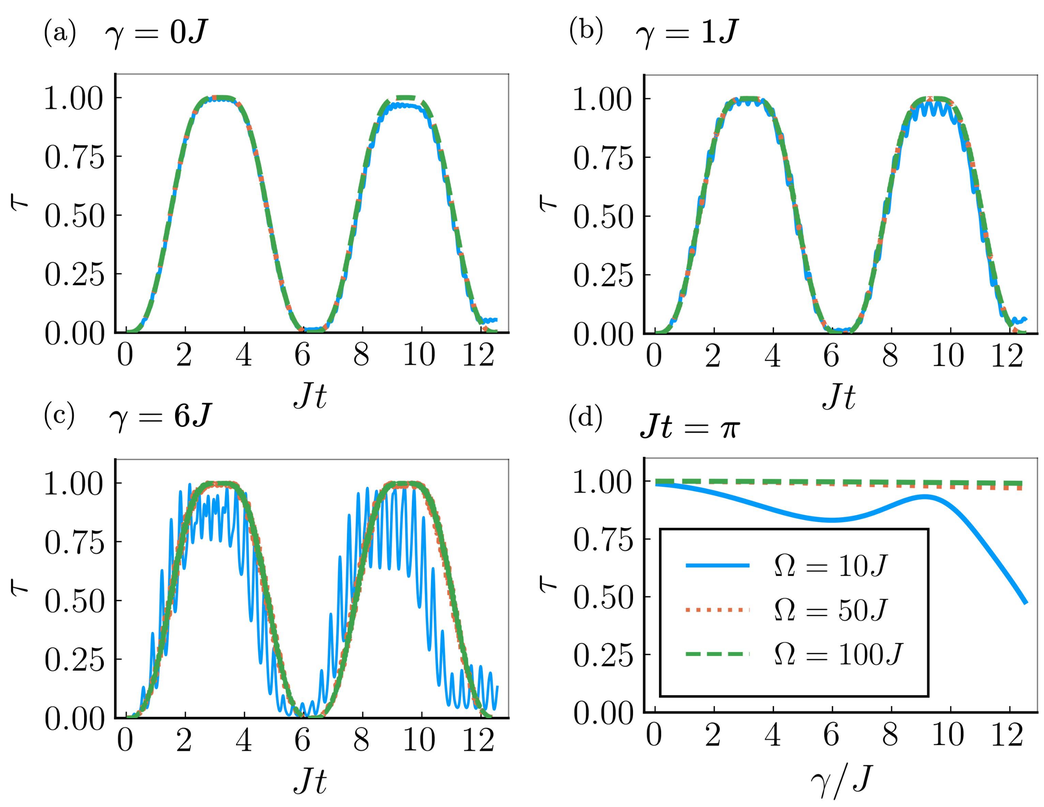
Method

→ Check **three tangle** $\tau = 1$
→ Check bipartite **entanglement entropy** $S = \log 2$
→ Local transforming output states to GHZ states and calculate **GHZ state fidelity** F

Consider **identical** qubits, coupling, driving, detuning → $\gamma, J, \Omega, \Delta$

GHZ state in Three qubits

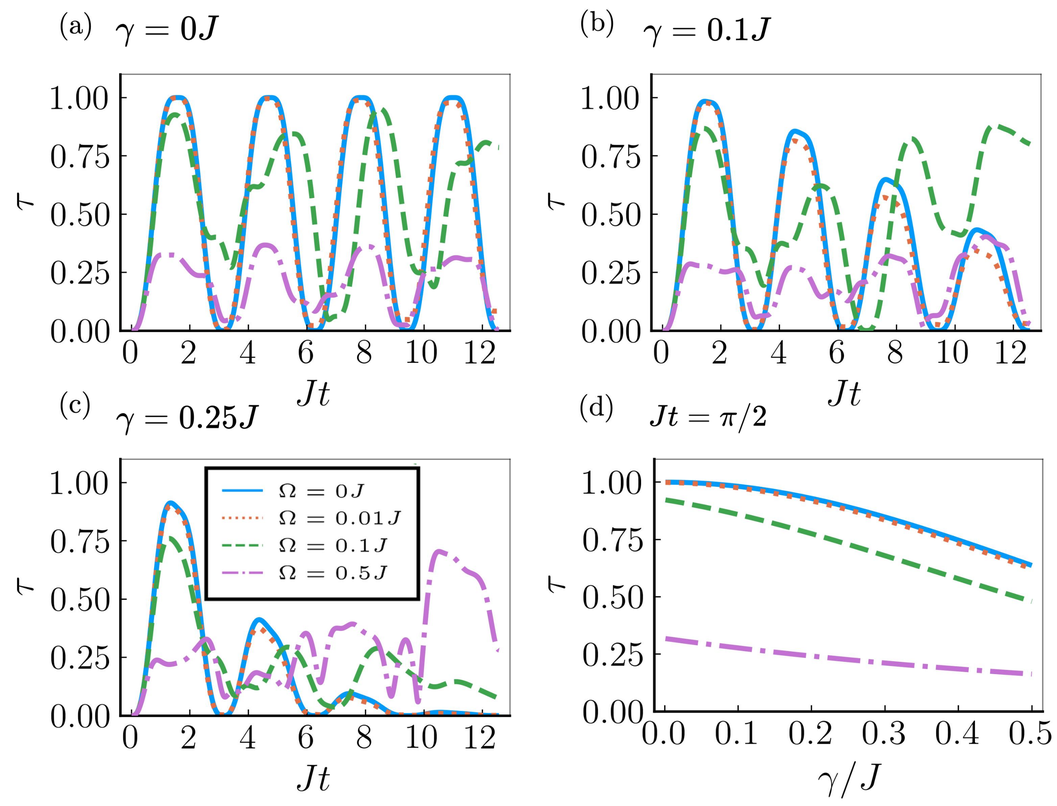
Strong driving regime with input **initial state:** $|\psi\rangle = (|f\rangle)^{\otimes 3}$



Results and Findings

→ **Three tangle** $\tau \approx 1$ when $Jt \approx (2n-1)\pi$ and reaches unity $(\tau \to 1)$ in the limit of $\Omega \to \infty$ and $Jt \to (2n-1)\pi$
→ Comparable Ω and γ, τ can be **restored**
→ Under strong **increased** Ω, τ can be robust against decay γ

Strong coupling regime with input **initial state**: $|\psi\rangle = \left(|f\rangle \pm e^{i\phi}|e\rangle\right)^{\otimes 3}$

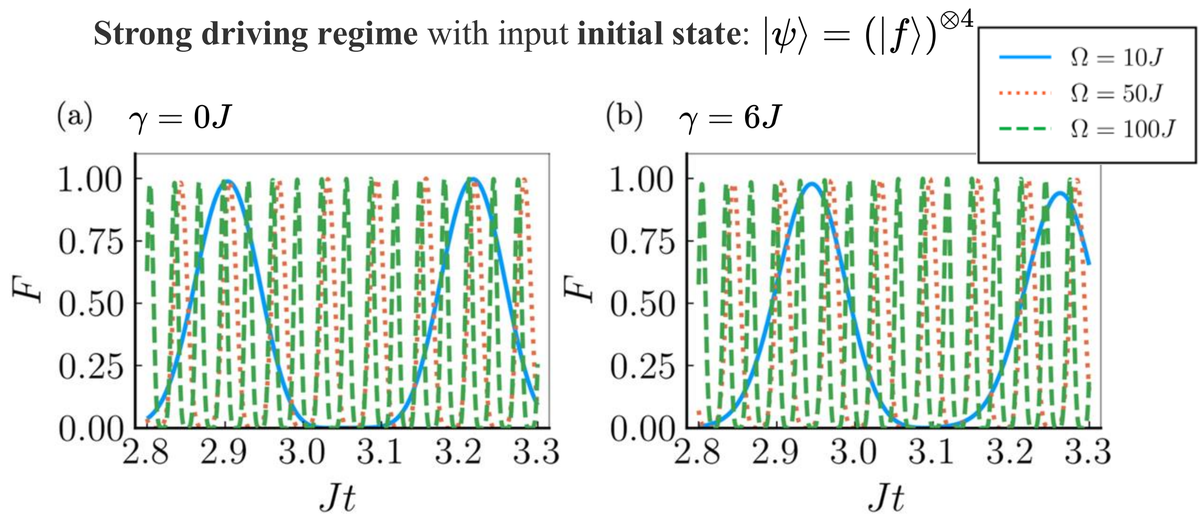


Results and Findings

→ **Three tangle** $\tau \approx 1$ when $Jt \approx (2n-1)\pi/2$ and reaches unity ($\tau = 1$) when no deacy $\gamma = 0$ and $Jt = (2n-1)\pi/2$
→ Under comparable Ω and γ, τ can be **restored at longer time**
→ Relative phase ϕ can be arbitrary for our entangling process

Extension to Four qubits

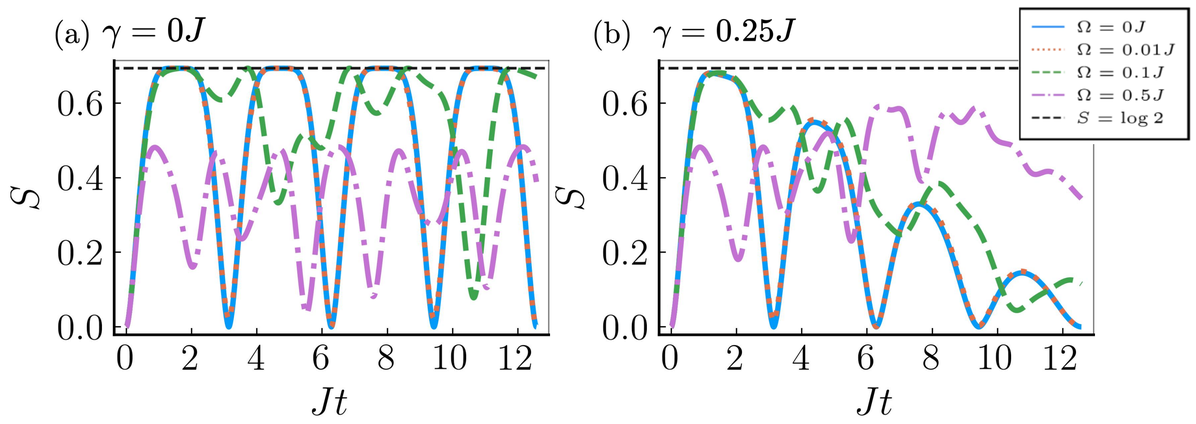
Strong driving regime with input **initial state**: $|\psi\rangle = (|f\rangle)^{\otimes 4}$



Results and Findings

→ **GHZ state fidelity** F between our output state and GHZ state (up to single qubit rotation $U_l = e^{-i\frac{3\pi}{4}}(Z(3\pi/2)\otimes I\otimes I\otimes I)$) oscillate at frequency proportional to Ω when $Jt \approx (2n-1)\pi$
→ There is a trade-off between **GHZ state fidelity** and **robustness over time**

Strong coupling regime with input **initial state**: $|\psi(0)\rangle = \left(|f\rangle + e^{i\phi}|e\rangle\right)^{\otimes 4}$



Results and Findings

→ **Entanglement entropy** $S \approx \log 2$ when $Jt \approx (2n-1)\pi/2$ and reaches unity ($S = \log 2$) when no deacy $\gamma = 0$ and $Jt = (2n-1)\pi/2$
→ Under comparable Ω and γ, τ can be **restored at longer time**
→ Possible **extensions** to higher qubits can be achieved using similar conditions